

Q1. Campus Router Health 360

DigiPlus IT Agentic AI Hackathon · 6-hour build · Thakur College of Engineering & Technology

1. Challenge

The college has issued 10,000 Wi-Fi routers to students and staff through its ISP partner. A few hundred perform badly — slow speeds, frequent disconnects, dead zones — but nobody knows which ones, where, or why. Complaints arrive one angry email at a time, and the IT team investigates each router from scratch.

2. Problem Statement

Build a router health monitor: compute a health score for every router from its metrics, show the worst performers, and provide an AI copilot that answers "why is router R-1042 performing badly?" with the actual cause, supporting evidence from the data, and a recommended fix.

3. Dataset

File	Contents
routers.csv	Router inventory: model, firmware version, building/room, user type.
metrics.csv	Hourly performance per router: speed, latency, packet loss, disconnects, devices, signal.
complaints.csv	User complaint tickets with free text, linked to a router.

4. Reference Flow (User Perspective)

Today — the pain:

1. A student in Hostel-B suffers Wi-Fi drops every evening; after days they email the IT helpdesk.
2. The mail sits in a shared inbox; an engineer eventually pulls logs for that one router manually.
3. There is no fleet view — no way to tell one bad router from a bad firmware batch.
4. The investigation repeats from scratch for the next complaint.

With your system:

1. The IT admin opens a dashboard and sees the worst-10 routers ranked by health score.
2. They click a router and see its key metrics and complaints together.
3. They ask the copilot "why is this router bad?" and get cause + evidence + one recommended fix.
4. They act before most affected users complain.

5. Expected Components in the Solution

Component	Expectation
Backend	Load the 3 CSVs; compute a per-router health score on a simple documented formula; serve 3 endpoints: rankings, router detail, copilot answer.
Frontend	One page: worst-10 table with scores, a detail panel (2 metric charts + complaints), and a copilot question box.
AI layer	Diagnosis grounded in the data: cause, the numbers supporting it, one fix (firmware / relocate / replace / user education).
Deployment	Live public URL (free tier) + public GitHub repo.
Bonus	Filters by building/firmware; health distribution chart; clustering routers into failure patterns.

Build the core completely before touching anything in the Bonus row. A small finished system beats a large broken one.

6. Validation Scenarios — How an Ideal System Behaves

- A router with sustained bad metrics ranks in the worst-10; a router with one bad hour does not.
- Asked about R-1042, the copilot cites real numbers from the dataset and recommends exactly one fix — never invented evidence.

- Asked about a healthy router, it says healthy and shows the numbers.
- A router with complaints but healthy metrics points to user education, not replacement.